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Statistical Modeling of Combinatorial Response Data

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Abstract

There is a rich literature for modeling binary and polychotomous responses. However, existing methods are inadequate for handling combinatorial responses, where each response is an integer array under additional constraints. Such data are increasingly common in modern applications, such as surveys collected under skip logic, event propagation on a network, and observed matching in ecology. Ignoring the combinatorial structure leads to biased estimation and prediction. The fundamental challenge is the lack of a link function that connects a linear or functional predictor with a probability respecting the combinatorial constraints. In this article, we propose a novel augmented likelihood that views combinatorial response as a deterministic transform of a continuous latent variable. We specify the transform as the maximizer of integer linear program, and characterize useful properties such as dual thresholding representation. When taking a Bayesian approach and considering a multivariate normal for the latent variable, our method becomes a direct generalization to the celebrated probit data augmentation, and enjoys straightforward computation via Markov chain Monte Carlo. We provide theoretical justification, including consistency and applicability, at an intersection between duality and probability. We demonstrate the effectiveness of our method through simulations and a data application on the seasonal matching between waterfowl.

Keywords: Combinatorial constraints, Integer linear programming, Strong duality, Data augmentation, Posterior consistency

1 Introduction

In statistical data analysis, combinatorial response data are becoming increasingly common. For example, in online surveys, since the user can terminate the survey more easily than in in-person questionnaires, survey designers often use skip logic to present only a subset of questions to each user. Depending on the user’s answer to each branching question, the vector record for each user’s answers has some elements marked as skipped (recorded as 0). Those zeros are often referred to as *structural zeros* in the literature (Manrique-Vallier and Reiter, 2014), as they appear not due to random mechanism, but due to the combinatorial constraints. Similarly, in ecological studies, seasonally monogamous animals like ducks form pair bonds gradually over the course of a year. Each observed matching graph at a given time is a subgraph of a bipartite graph, which describes the possible pair bonds (for example, between male and female, belonging to the same species), and each node in the matching graph can have at most one edge, corresponding to the pairing constraint.

In applications such as above, the multivariate data are combinatorial since they are not only discrete, but also subject to additional constraints. That means the response space is in fact much smaller than the product space of all possible marginal outcomes. Under a likelihood approach, the probability of those outcomes not satisfying the constraints is effectively zero. It is of great interest to characterize the probability distribution of such data and develop regression models characterizing their relationship with covariates.

The existing strategies to model such combinatorial response can be put into two categories. First, one may enumerate all possible outcomes in the combinatorial space and reindex each response as a univariate multinomial outcome instead. Such a strategy is suitable when the response is in a low dimension, and the number of possibilities in each dimension is finite. On the other hand, such a reindexing strategy becomes impractical as the dimension increases exponentially (Baptista and Poloczek, 2018), and reindexing may lead to ignoring ordinal information between two combinatorial outcomes.

Second, one may simply ignore the constraints and model each data point as an array of discrete outcomes. To elucidate the relation between two likelihoods, $\mathcal{L}(y; \theta)$ upholding and $\bar{\mathcal{L}}(y; \theta)$ ignoring the constraints, with y the data and θ the parameter, we can think of $\mathcal{L}(y; \theta)$ as a conditional probability given that discrete vector falling inside the constrained space denoted by $\mathcal{Y}$ hence $\mathcal{L}(y; \theta) = \bar{\mathcal{L}}(y; \theta)1(y \in \mathcal{Y}) / P(y \in \mathcal{Y}; \theta)$. Clearly, omitting $P(y \in \mathcal{Y}; \theta)$ results in a biased estimation of θ , which leads to inconsistency as we demonstrate later. Similarly, as has long been recognized in the literature on log-linear and latent class models, failure to properly account for structural zeros introduces bias into the estimated distribution (Goodman, 1968; Fienberg, 1972; Bishop et al., 1975; Vermunt, 1997); for a more recent discussion, see Manrique-Vallier and Reiter (2014); Carota et al. (2015).

To appropriately model combinatorial data under constraints, we propose a generative approach that first samples a latent vector from an unconstrained continuous distribution, and then deterministically maps it to a combinatorial outcome in $\mathcal{Y}$. This continuous latent representation, known as an *embedding*, builds on foundational work by Albert and Chib (1993) who showed

that binary outcomes can be modeled by thresholding a univariate normal at zero. Their key insight was that marginalizing out this latent variable yields probabilities equivalent to those from a generalized linear model (GLM) with a link function. However, the embedding framework offers greater flexibility than traditional GLMs.

Several extensions demonstrate the versatility of latent embeddings. Ordinal data can be handled by introducing multiple thresholds to partition the latent space into ordered categories (Albert and Chib, 1993). For network data with binary symmetric adjacency matrices, low-rank matrix embeddings with normal noise have been employed (Hoff et al., 2002). Rank data correlations can be modeled using transformed multivariate normal vectors (Hoff, 2007). In spatial statistics, spatial dependence between categorical data can be captured through latent Gaussian processes (Rue et al., 2009). These examples highlight how embeddings can naturally incorporate complex dependencies and constraints.

Our approach stems from reinterpreting the zero-thresholding transform of Albert and Chib (1993) as the solution to a univariate integer linear program. We extend this insight by generalizing the transform to a multivariate integer linear program, which naturally accommodates combinatorial constraints. Since we observe the combinatorial data, our objective is not to solve an optimization problem, but to determine the continuous embedding that corresponds to the observed integer solution. We demonstrate that this continuous embedding is analytically tractable through strong duality, and develop a Metropolis-Hastings-Within-Gibbs sampler with excellent computational performance.

2 Method

Our focus is on regression with d -dimensional integer response y_i in the space $\{z \in \mathbb{Z}^d : Az \leq b\}$, where A is an $m \times d$ matrix, b is a m -element vector, $\mathbb{Z}$ is the integer set, and $\leq$ applies elementwise. The coefficients in A and b are pre-determined (up to common scaling) by known constraints for the data. To be concrete, we now list a few simple examples, and will introduce a few more complex constraints in Section 2.4.

First, for binary constraint $y_i \in \{0,1\}^d$, $Ay_i \leq b$ corresponds to $0 \leq y_{i,j} \leq 1$ with $A = [I, -I]$ and $b = [1_d, 0_d]$. Second, for equality constraint $a_k^\top y_i = b_k$, such as zero-contrast $y_{i,1} + y_{i,2} - y_{i,3} - y_{i,4} = 0$, $Ay_i \leq b$ contains two inequalities $a_k^\top y_i \leq b_k$ and $-a_k^\top y_i \leq -b_k$. Third, for ordered or partially ordered integer data, such as non-decreasing sequence $y_{i,1} \leq y_{i,2} \leq \dots \leq y_{i,d}$, $Az \leq b$ contains inequality $y_{i,j} - y_{i,k} \leq 0$ for suitable (j, k) .

2.1 Data augmentation via integer linear program

In order to facilitate the introduction of our data augmentation (DA) strategy, we first review the DA for the probit model for simple binary data $y_i \in \{0,1\}$. Albert and Chib (1993) shows that for $y_i \sim \text{Bernoulli}[\Phi(\mu_i)]$, with $\mu_i \in \mathbb{R}$ (such as in the form of linear predictor) and Φ the

distribution function for standard normal, there is a latent normal $\zeta_i \sim N(\mu_i, 1)$ which has $P(\zeta_i > 0) = \Phi(\mu_i)$, $P(\zeta_i < 0) = 1 - \Phi(\mu_i)$. In another words, ζ_i is a latent variable for y_i , via thresholding at zero: $y_i = 1(\zeta_i > 0)$. We now provide a new perspective using integer linear program, with coefficient $\zeta_i \neq 0$ almost surely:

$$y_i = \arg \max_{z \in \{0,1\}} \zeta_i z.$$

It is not hard to see that the above has the same solution: $y_i = 1$ if $\zeta_i > 0$, and $y_i = 0$ if $\zeta_i < 0$. Taking this idea further, we now consider $y_i \in \mathbb{Z}^d$ as the solution to multivariate integer linear program:

$$y_i = \arg \max_z \zeta_i^\top z, \quad \text{subject to } z \in \mathbb{Z}^d, Az \leq b. \quad (1)$$

We denote the continuous polyhedron as $\mathcal{P} = \{z^* : Az^* \leq b\}$; hence, the feasible region of the above problem is equivalent to $\mathbb{Z}^d \cap \mathcal{P}$. The parameterization (1) enables us to model combinatorial data y_i under affine constraints $Ay_i \leq b$.

Shopper's selection problem: To make the integer linear program more intuitive and interpretable, we describe a thought experiment: shopper's selection. Imagine a shopper i is buying several items from a store, the store has d items, each item has a utility score $\zeta_{i,j}$ that represents how desirable item j is to shopper i ($\zeta_{i,j} > 0$ desirable, and $\zeta_{i,j} < 0$ undesirable), and $y_i \in \mathbb{Z}^d$ records the units of each item the shopper selects to buy. The shopper has a set of constraints on the selection. For examples, the shopper may have a budget constraint that the total cost is at most b_1 ; the shopper may also have a need constraint that at least b_2 items need to be selected under some category; last but not least, the shopper cannot select negative units of items. Clearly, the integer linear program (1) corresponds to the shopper's maximization of total utility score subject to the constraints.

Since we observe y_i only, we need to model ζ_i . In our approach, any continuous distribution on $\mathbb{R}^d$ can be used for ζ_i . When ζ_i is from a continuous distribution, $\zeta_i^\top (z_1 - z_2) = 0$ has probability zero for any two distinct $z_1, z_2 \in \mathbb{Z}^d \cap \mathcal{P}$, so the solution to the integer linear program (1) is unique almost surely. Therefore, for the maximum value $\zeta_i^\top \hat{z}$, we know $\hat{z}$ is unique almost surely with respect to the continuous distribution of ζ_i .

The almost sure unique solution means that we can consider (1) as a mapping:

$$T: \mathbb{R}^d \setminus C \mapsto \mathbb{Z}^d, \quad y_i = T(\zeta_i) = \arg \max_{z \in \mathbb{Z}^d \cap \mathcal{P}} \zeta_i^\top z, \quad (2)$$

where C is the set that does not lead to a unique solution, and C has measure zero under the distribution of ζ_i . The maximizer may not have a closed-form and hence T is an *implicit function* equivalent to $y_i : \zeta_i^\top y_i = \max_{z \in \mathbb{Z}^d \cap \mathcal{P}} \zeta_i^\top z$. In the rest of the article, all of the statements about $T(\zeta_i)$ are meant in the almost sure sense.

Taking a generative view of continuous $\zeta_i \sim \mathcal{F}(\cdot; \mu_i)$ and transform $y_i = T(\zeta_i)$, we have an augmented likelihood:

$$\mathcal{L}(y_i, \zeta_i; \mu_i) = \mathcal{F}(\zeta_i; \mu_i) \mathbb{1}(y_i = \arg \max_{z \in \mathbb{Z}^d \cap \mathcal{P}} \zeta_i^\top z). \quad (3)$$

Marginally, we can integrate over ζ_i and find the probability of y_i taking one specific combinatorial value e :

$$P(y_i = e; \mu_i) = \int_{T^{-1}(e)} \mathcal{F}(\zeta_i; \mu_i) d\zeta_i = \int_{\mathbb{R}^d} \mathcal{F}(\zeta_i; \mu_i) \mathbb{1}(\zeta_i^\top e = \max_{z \in \mathbb{R}^d \cap \mathcal{P}} \zeta_i^\top z) d\zeta_i, \quad (4)$$

where $T^{-1}(e) := \{\zeta_i \in \mathbb{R}^d : T(\zeta_i) = e\}$ is the pre-image of e . Following the nomenclature in GLM, we refer to (4) as the Inverse Integer Linear Program (IILP)-link, since it is associated with the link function between μ_i and $\mathbb{E}(y_i | \mu_i)$.

We now provide an illustrative toy example. Consider two-dimensional $y_i \in \{0, 1\}^2$ with $y_{i,1} + y_{i,2} \leq 1$. This integer linear program has a closed-form solution $y_i = (0, 0)$, when $\zeta_{i,1} < 0, \zeta_{i,2} < 0$; $(0, 1)$, when $\zeta_{i,1} < \zeta_{i,2}, \zeta_{i,2} > 0$; $(1, 0)$, when $\zeta_{i,1} > \zeta_{i,2}, \zeta_{i,1} > 0$. If we further specify $\zeta_i \sim \mathcal{N}[(\mu_0, \mu_0)^\top, I_2]$, then we see the three possible outcomes have likelihood valued at $\Phi^2(-\mu_0)$, $0.5[1 - \Phi^2(-\mu_0)]$ and $0.5[1 - \Phi^2(-\mu_0)]$, respectively.

While ζ_i is interpretable via the shopper's selection problem, we treat it as an auxiliary variable as our interest is on the marginal distribution of combinatorial response. There may be alternative parameterizations for the same marginal probability. For example, one could consider

$$p(y_i; \gamma) = m^{-1}(\gamma) \mathbb{1}(y_{i,1} + y_{i,2} \leq 1) \prod_{i=1}^2 \Phi(\gamma)^{y_{i,j}} \Phi(-\gamma)^{1-y_{i,j}}, \text{ the distribution of two independent}$$

Bernoulli random variables conditioned on their sum being less than or equal to 1, with $m(\gamma)$ being the normalization constant. It is not hard to see its equivalence to our model via $\Phi^2(-\mu_0) = \Phi(-\gamma) / [2 - \Phi(-\gamma)]$.

On the other hand, a major advantage of our proposed approach over the alternative is that the augmented likelihood does not involve $m^{-1}(\cdot)$, which may become intractable as the number of constraints increases. Although there are a plethora of algorithms for dealing with intractable normalizing constants, such as exchange algorithms (Murray et al., 2006; Møller et al., 2006; Liang et al., 2016) and pseudo-marginal Metropolis-Hastings (Andrieu and Roberts, 2009; Lee et

al., 2019), our model without intractable normalizing constant enjoys algorithmic simplicity and good scalability to relatively high dimension.

Remark 1 .

The shopper's selection example endows ζ_i with a concrete meaning; in general, when ζ_i is not directly meaningful, we can specify an appropriate model for y_i following the principles of generalized linear models (Agresti, 2018, Chapter 3.1) by breaking the model into two parts: a systematic component μ_i that depends on the covariates and a random component that perturbs μ_i into ζ_i , which is in turn transformed into the response $y_i = T(\zeta_i)$. To provide an analogy, in univariate binary regression, one could add either standard logistic error or standard normal error to $\mu_i \in \mathbb{R}$, then threshold at 0, leading to logistic regression or probit regression, respectively. In both cases, ζ_i may not be interpretable but simply as a noisy version of μ_i that characterizes the stochasticity from the deterministic part μ_i to the response y_i . In our framework, we use $\zeta_i \sim \mathcal{F}(\cdot; \mu_i)$ for the same purpose. In particular, we choose the standard multivariate normal error, i.e., $\zeta_i \sim N(\mu_i, I)$, which is motivated by the computational convenience.

Our next task is to find out $T^{-1}(e)$, the appropriate set of input ζ_i for each possible outcome e . In the next two subsections, we first introduce a condition on $\mathcal{P}$, known as *integrality*, that leads to an equivalent solution between integer linear program and continuous linear program, then we describe how to find $T^{-1}(e)$ when $\mathcal{P}$ is integral.

2.2 Integral polyhedron and image of integer linear program

The feasible region of the integer linear program (1) is an integer set. We now consider a relaxation that extends the feasible region to a polyhedron $\mathcal{P} = \{z \in \mathbb{R}^d : Az \leq b\}$, which makes the problem a continuous linear program:

$$\max_{z \in \mathbb{R}^d} \zeta_i^\top z, \text{ subject to } z \in \mathcal{P}. \quad (5)$$

A useful property of continuous linear program is the vertex optimality: if (i) $\mathcal{P}$ has at least one vertex (z that cannot be written as $\lambda z_1 + (1-\lambda)z_2$ for any distinct $z_1, z_2 \in \mathcal{P}$ and $\lambda \in (0,1)$), and (ii) $\max \zeta_i^\top z < \infty$, then at least one of the optimal solutions is on a vertex.

When $\mathcal{P}$ has all vertices integer-valued, $\mathcal{P}$ is referred to as an *integral polyhedron*, or simply, that $\mathcal{P}$ is integral. Therefore, under the integrality of $\mathcal{P}$, the continuous linear program (5) has at least one solution being integer-valued. Since $\max_{z \in \mathbb{Z}^d \cap \mathcal{P}} \zeta_i^\top z \leq \max_{z \in \mathbb{R}^d \cap \mathcal{P}} \zeta_i^\top z$ becomes equality, the solution of (1) belongs to the solution set of (5).

We now show a stronger result: as long as $\mathcal{P}$ has more than one vertex, the solution of (5) is unique almost surely, hence the solution of (5) being guaranteed to be the same as the one of (1) with probability one.

Theorem 1 .

When ζ_i is from a continuous distribution, for an integral $\mathcal{P}$ with more than one vertex, the solution to the continuous linear program (5) is integer-valued and unique almost surely.

We provide the proofs of theorems in the Supplementary Materials S1. We will describe the sufficient conditions on A and b that guarantee the integrality of $\mathcal{P}$; for now, we assume integral $\mathcal{P}$ throughout the rest of the article. Figure 1(a) illustrates how a linear program with integral polyhedron transforms continuous ζ_i into an integer-valued y_i .

The above uniqueness result also means that any non-vertex point in $\mathcal{P}$ is not optimal. Therefore, one should exercise caution that the range of T may not cover some combinatorial outcome in $\mathbb{Z}^d \cap \mathcal{P}$. Fortunately, when y_i is binary, we know any $z \in \{0,1\}^d : Az \leq b$ must be a vertex (as it cannot be written as a nontrivial convex combination of two other binary points). For ease of discussion and with some loss of generality, we reduce our focus to binary linear program $y_i \in \{0,1\}^d \cap \mathcal{P}$, such that

$$y_i = \arg \max_z \zeta_i^\top z, \text{ subject to } z \in [0,1]^d, Az \leq b. \quad (6)$$

In general, the integer linear program may lack a closed-form solution. Nevertheless, we show that given y_i , we can find the pre-image $T^{-1}(y_i)$ using the dual form.

2.3 Dual thresholding representation

For ease of notation, we now omit the subscript i and use $\eta = \zeta_i$ and $z = y_i$, and consider equivalent primal problem $\min_z (-\eta^\top z)$ subject to $z \in [0,1]^d, Az \leq b$.

Using $u \geq 0, v \geq 0, w \geq 0$, we set up the Lagrangian function

$$\begin{aligned} L(z, u, v, w) &= -\eta^\top z + u^\top (Az - b) + v^\top (z - 1) + w^\top (-z) \\ &= (-\eta^\top + u^\top A + v^\top - w^\top)z - u^\top b - v^\top 1. \end{aligned}$$

Minimizing over z leads to the dual problem:

$$\min_{u, v, w} u^\top b + v^\top 1$$

$$\text{subject to } -\eta + A^\top u + v = w \geq 0, u \geq 0, v \geq 0.$$

It is not hard to see the optimal $\hat{v} = (\eta - A^\top u)_+$, with $(x)_+ = x$ if $x > 0$ and 0 otherwise, hence simplifying the dual loss to $u^\top b + 1^\top (\eta - A^\top u)_+$.

We use complementary slackness and strong duality to reveal a geometric characterization between η and $\hat{z}$. First, on complementary slackness, we have three possible cases:

- Case (i): when $\eta_j > A_j^\top \hat{u}$, the optimal $\hat{v}_j = (\eta_j - A_j^\top \hat{u}) > 0$, so $\hat{w}_j = 0$ and $\hat{z}_j = 1$;
- Case (ii): when $\eta_j < A_j^\top \hat{u}$, the optimal $\hat{v}_j = 0$ and $\hat{w}_j = A_j^\top \hat{u} - \eta_j > 0$ hence $\hat{z}_j = 0$;
- Case (iii): when $\eta_j = A_j^\top \hat{u}$, the optimal $\hat{w}_j = \hat{v}_j = 0$. One may find $\hat{z}_j$ via the other complementary slackness conditions $\hat{u}_l (A\hat{z} - b)_l = 0$ for $l = 1, \dots, m$.

Second, the linear program enjoys strong duality

$$\max_{z \in [0,1]^d : Az \leq b} \eta^\top z = \min_{u \geq 0} u^\top b + 1^\top (\eta - A^\top u)_+,$$

which means that a feasible (z, u) achieves $\eta^\top z = u^\top b + 1^\top (\eta - A^\top u)_+$ if and only if (z, u) are the primal and dual optimal, respectively. Combining the above, we have a useful characterization between η and u .

Theorem 2 .

For a given $\tilde{z}$ in the primal feasible region, consider a set $C_{\tilde{z}}$ of (η, u) with $\eta \in \mathbb{R}^d, u \geq 0$ that satisfy,

- $u^\top (A\tilde{z} - b) = 0$,
- $\eta_j \geq A_j^\top u$ for $j : \tilde{z}_j = 1$, and $\eta_j \leq A_j^\top u$ for $j : \tilde{z}_j = 0$,

Then $\tilde{z}$ is a solution of $\max_{z \in [0,1]^d : Az \leq b} \eta^\top z$ if and only if $(\eta, u) \in C_{\tilde{z}}$.

We now switch to notation y_i and ζ_i , and index u as $u^{(i)}$. Theorem 2 gives a concrete characterization of the pre-image of the projection T defined in (2). For any y_i in the combinatorial space, the pre-image $T^{-1}(y_i)$ is given by

$$T^{-1}(y_i) = \bigcup_{u^{(i)} \geq 0 : u_k^{(i)} = 0 \text{ if } (Ay_i - b)_k < 0} \left\{ \zeta_i \in \mathbb{R}^d : \zeta_{i,j} \geq A_j^\top u^{(i)} \text{ if } y_{ij} = 1, \zeta_{i,j} < A_j^\top u^{(i)} \text{ if } y_{i,j} = 0 \right\}.$$

Equivalently, we have a simple interpretation: there exists some $u^{(i)} \geq 0$ such that

$$y_{i,j} = \mathbb{1}(\zeta_{i,j} \geq A_{\cdot j}^\top u^{(i)}). \quad (7)$$

We refer to (2.3) as the *dual thresholding representation*, and illustrate this in Figure 1(b). When there are no constraints on the binary vector y_i , we recover the zero thresholding $y_i = \mathbb{1}(\zeta_i > 0)$ as in Albert and Chib (1993).

To be clear, the augmented likelihood $\mathcal{L}(y_i, \zeta_i; \mu_i)$ in (3) describes the distribution of y_i and ζ_i , and the dual form (7) should be understood as the existence of a separating set:

$$\begin{aligned} \mathcal{U}(y_i, \zeta_i) = \{u : \zeta_{i,j} \geq A_{\cdot j}^\top u^{(i)} : y_{i,j} = 1, \zeta_{i,j} \leq A_{\cdot j}^\top u^{(i)} : y_{i,j} = 0, \text{ for } j = 1 \dots d, \\ u_k = 0 \text{ for } k : (A y_i - b)_k < 0, u_k \geq 0 \text{ for } k : (A y_i - b)_k = 0\}. \end{aligned}$$

That is, $y_i = \arg\max_{z \in \{0,1\}^d \cap \mathcal{P}} \zeta_i^\top z \Leftrightarrow \mathcal{U}(y_i, \zeta_i) \neq \emptyset$. To enforce $\mathcal{U}(y_i, \zeta_i) \neq \emptyset$ in the posterior

computation, we use another latent variable $u^{(i)} \sim \Pi_u(\cdot | y_i, \zeta_i)$ supported on $\mathcal{U}(y_i, \zeta_i)$:

$$\begin{aligned} \mathcal{L}(y_i, \zeta_i; \mu_i) \Pi_u(u^{(i)} | y_i, \zeta_i) = \mathcal{F}(\zeta_i; \mu_i) \prod_{j=1}^d \{\mathbb{1}(y_{i,j} = 1, \zeta_{i,j} > A_{\cdot j}^\top u^{(i)}) + \mathbb{1}(y_{i,j} = 0, \zeta_{i,j} < A_{\cdot j}^\top u^{(i)})\} \quad (7) \\ \mathbb{1}[u^{(i)} \in \mathcal{U}(y_i, \zeta_i), \mathcal{U}(y_i, \zeta_i) \neq \emptyset] \cdot \Pi_u(u^{(i)} | y_i, \zeta_i), \quad (8) \end{aligned}$$

where $\Pi_u(u^{(i)} | y_i, \zeta_i)$ is an arbitrary proper density, and we have replaced $\geq$ by $>$ in $\mathcal{U}(y_i, \zeta_i)$ since the equality event has zero probability when ζ_i and $u^{(i)}$ are continuous variables.

2.4 Applicability of data augmentation

For our data augmentation to work, one key condition is that the polyhedron $\mathcal{P} = \{z \in \mathbb{R}^d : Az \leq b\}$ is integral. Now we present a few useful results from the combinatorial optimization literature for checking this condition, and we illustrate the condition with a few examples commonly seen in applications. The results in this section apply to general integer z that is not necessarily binary; whereas the binary setting can be viewed as having constraint $0 \leq z \leq 1$ in addition to $\tilde{A}z \leq \tilde{b}$, so we view $A = [\tilde{A}, I, -I]$ and $b = [\tilde{b}, 1, 0]$.

A useful sufficient (though not necessary) condition for a polyhedron to be integral is total unimodularity. A matrix A is said to be totally unimodular (TUM) if every square submatrix of A has determinant in $\{-1, 0, 1\}$. Then we have the following (Schrijver, 1998):

Theorem 3 .

If A is TUM, then for any $b \in \mathbb{Z}^m$, the polyhedron $\mathcal{P} = \{z \in \mathbb{R}^d : Az \leq b\}$ is an integral polyhedron.

The converse holds as well (Hoffman and Kruskal, 2010), that if $\mathcal{P}$ is integral for all $b \in \mathbb{Z}^m$, then A is TUM. It is not hard to see that $A_{i,j} \in \{-1, 0, 1\}$. The total unimodularity can be easily checked via software, such as `lintools` package in R. On the other hand, we list a few detailed properties that could be useful for model extension.

1. If A is TUM, then both $-A$ and $A^\top$ are TUM.
2. If A is TUM, then $[A, I]$ is TUM.
3. If A is TUM, then $[A, -A]$ is TUM.
4. If $A_{i,j} \in \{-1, 0, 1\}$, each column of A has at most one -1 and at most one 1, then A is TUM.
5. If $A_{i,j} \in \{-1, 0, 1\}$, each column of A has at most two non-zeros, and there exists a row index subset $S \subseteq [m]$: $\sum_{i \in S} A_{i,j} = \sum_{i \notin S} A_{i,j}$ for all j , then A is TUM.

The proof can be found in Wolsey and Nemhauser (1999). We give a few useful examples.

Example 1 (Binary constraint) In our focused binary linear program $z \in [0, 1]^d$, using Properties 1 and 2, and the fact that multiplying a row of square submatrix by -1 keeps the determinant in $\{-1, 0, 1\}$, we see $A = [\tilde{A}, I, -I]$ is TUM if $\tilde{A}$ is TUM.

Example 2 (Equality) For equality constraints $\tilde{A}z = b$ with a TUM $\tilde{A}$, we can represent equality by $[\tilde{A}, -\tilde{A}]z \leq [b, -b]$, hence we see the polyhedron is integral via Property 3.

Example 3 (Partial ordering) For partial order constraints $z_j \leq z_k$, for a set of pair indices (j, k) , we can represent them by $A_{i,j} = 1$, $A_{i,k} = -1$ and $A_{i,l} = 0$ $l \neq j$ or k , with $b = 0$, hence Property 4 leads to A being TUM.

Example 4 (Matching) Suppose we have a bipartite graph $G = (V, E_G)$, with V the node set divided into two subsets V_1 and V_2 , and E_G the edges formed only across V_1 and V_2 . A matching is a subgraph $M = (V, E_M)$ with $E_M \subseteq E_G$, such that each node has zero or one edge connecting to another node. We can use $z \in \{0, 1\}^{|E_G|}$ to represent if the edge of E_G is chosen in E_M . Using a matrix $\tilde{A}$ for G , with $\tilde{A}_{v,e} = 1$ if the e -th edge in E_G is incident to node v , otherwise $\tilde{A}_{v,e} = 0$, we

see that $\tilde{A}$ is TUM using Property 5 with $S = V_1$. The matching representation z is associated with a TUM $A = [\tilde{A}, -I]$ and integer $b = [1, 0]$.

The above could be extended to perfect matching where each node has exactly one edge, or $\tilde{b}$ -matching where each node's edge upper bound a varying integer.

Example 5 (Flow) Consider a flow network with V the node set and E the directed edge set, with s the source node and t the sink node. The flow on the network is represented by $z_e \geq 0$ for directed edge $e \in E$, subject to capacity constraint $z_e \leq \tilde{b}_e$, and flow conservation for each node $i \in V \setminus \{s, t\}$, $\sum_{e \in E: e=(i \rightarrow j)} z_e = \sum_{e \in E: e=(k \rightarrow i)} z_e$. We see that the incidence matrix $\tilde{A}$ for directed graph is TUM, where $\tilde{A}_{i,e} = 1$, $\tilde{A}_{j,e} = -1$ for each $e = (i \rightarrow j)$, and $\tilde{A}_{l,e} = 0$ for $l \neq i$ or j . The flow network representation vector z is associated with a TUM matrix $A = [\tilde{A}, -\tilde{A}, I, -I]$ and $b = [0, 0, \tilde{b}, 0]$. If we have all capacity constraints $\tilde{b} \in \mathbb{Z}^{|E|}$, then we know the produced flow under a linear program is integer-valued.

We want to clarify that TUM is only one of the useful sufficient conditions for the integrality of a polyhedron. Other conditions often used in practice include the total dual integrality condition and Edmonds-Giles' theorem (Edmonds and Giles, 1977), where A need not be TUM. We refer the readers to Schrijver (1998) for a comprehensive review.

3 Sampling algorithms for Bayesian regression with combinatorial response

We now focus on a Bayesian regression model with combinatorial response $y_i \in \{0, 1\}^d$ and linear predictor $\mu_i = \beta x_i$, for $x_i \in \mathbb{R}^p$ and $\beta \in \mathbb{R}^{d \times p}$. We assign a matrix-normal prior $\beta \sim \text{Mat-N}(0, I_d, I_p \tau)$ with $\tau > 0$ a hyper-parameter, and use $\zeta_i \sim N(\mu_i, I_d)$. We have the augmented likelihood independent for $i = 1, \dots, n$:

$$\mathcal{L}(y_i, \zeta_i; \beta x_i) \propto \prod_{j=1}^d \{1(y_{i,j} = 1, \zeta_{i,j} > (UA)_{i,j}) + 1(y_{i,j} = 0, \zeta_{i,j} < (UA)_{i,j})\} \\ \times \exp(-0.5 \|\zeta_i - \beta x_i\|_2^2) \mathbb{I}[u^{(i)} \in \mathcal{U}(y_i, \zeta_i), \mathcal{U}(y_i, \zeta_i) \neq \emptyset],$$

For simpler notation, matrix $U \in \mathbb{R}^{n \times m}$ has (i, j) -th entry $u_j^{(i)}$, hence $(UA)_{i,j} = A_j^\top u^{(i)}$.

We now present a Metropolis-Hastings-within-Gibbs sampler. Continuing from our discussion of (8), we now view $u^{(i)}$ as a random variable and specify $\Pi_u(u^{(i)} | y_i, \zeta_i)$. We assign exponential kernel for those $u_k^{(i)} > 0$, leading to

$$\Pi_u(u^{(i)} | y_i, \zeta_i) = \frac{\prod_{k: (Ay_i - b)_k = 0} \exp(-u_k^{(i)})}{c[\mathcal{U}(y_i, \zeta_i)]}, \quad (9)$$

where $c[\mathcal{U}(y_i, \zeta_i)]$ is the normalizing constant. Since $c[\mathcal{U}(y_i, \zeta_i)]$ depends on ζ_i , whose value is likely intractable, we propose an accept–reject step to bypass its computation. For two sets $\mathcal{U}(y_i, \zeta_i)$ and $\mathcal{U}(y_i, \zeta_i^*)$, consider a third set $\mathcal{U}(y_i, \tilde{\zeta}_i)$, with $\tilde{\zeta}_{i,j} = \max(\zeta_{i,j}, \zeta_{i,j}^*)$ if $y_{i,j} = 1$ and $\tilde{\zeta}_{i,j} = \min(\zeta_{i,j}, \zeta_{i,j}^*)$ if $y_{i,j} = 0$. Clearly, $\mathcal{U}(y_i, \zeta_i)$ and $\mathcal{U}(y_i, \zeta_i^*)$ are subsets of $\mathcal{U}(y_i, \tilde{\zeta}_i)$.

Consider the event of drawing $u^{(i)*} \sim \Pi_u(\cdot | y_i, \tilde{\zeta}_i)$ in $\mathcal{U}(y_i, \tilde{\zeta}_i)$, and having $u^{(i)*}$ falling in the polyhedron $\mathcal{U}(y_i, \zeta_i)$, it has probability:

$$P[u^{(i)*} \in \mathcal{U}(y_i, \zeta_i)] = \frac{c[\mathcal{U}(y_i, \zeta_i)]}{c[\mathcal{U}(y_i, \tilde{\zeta}_i)]}.$$

Therefore, having an additional acceptance step using the above probability is essential in order to cancel out $c[\mathcal{U}(y_i, \zeta_i)]$ in the acceptance ratio. Similar scenarios of dealing with intractable normalizing constants were considered by Møller et al. (2006); Lee (2011). In theory, any proper distribution having support $\mathcal{U}(y, \zeta)$ is a valid distribution choice for $u^{(i)}$. In the Supplementary Materials S6, we show a simulation using alternative half-Gaussian kernel for $u^{(i)}$, and find no empirical difference from exponential in mixing performance.

Each MCMC iteration of our algorithm consists of the following steps:

- Update ζ_i for each i :

1. Generate $u^{(i)}$ from $\Pi(u^{(i)} | y_i, \zeta_i)$ as in (9). Propose ζ_i^* based on truncated normal over

elements $j \in \{1, \dots, d\}$: $\zeta_{i,j}^* \sim \begin{cases} \text{TN}(\beta_j^\top x_i, 1) \mathbb{I}[\cdot > (UA)_{i,j}], & \text{if } y_{i,j} = 1; \\ \text{TN}(\beta_j^\top x_i, 1) \mathbb{I}[\cdot < (UA)_{i,j}], & \text{if } y_{i,j} = 0. \end{cases}$

2. Based on $\mathcal{U}(y_i, \zeta_i)$ and $\mathcal{U}(y_i, \zeta_i^*)$, find $\mathcal{U}(y_i, \tilde{\zeta}_i)$, then draw another $u^{(i)*}$ from $\Pi(u^{(i)} | y_i, \tilde{\zeta}_i)$.

3. If $u^{(i)*} \in \mathcal{U}(y_i, \zeta_i)$, accept ζ_i^* as the updated value of ζ_i . Otherwise, keep the current value of ζ_i .

- Update β from matrix normal (where $X \in \mathbb{R}^{p \times n}$, $\zeta \in \mathbb{R}^{n \times d}$)

$(\beta \mid \cdot) \sim \text{Mat-N}[\zeta^\top X^\top (XX^\top + I_p / \tau)^{-1}, I_d, (XX^\top + I_p / \tau)^{-1}]$.

To explain the algorithm, we examine the first step as a Metropolis-Hastings step. On the transition from ζ_i to ζ_i^* , we have the joint probability kernel of the current ζ_i and proposed ζ_i^* , marginalized over $u^{(i)}$:

$$\begin{aligned} & \mathcal{L}(y_i, \zeta_i; \beta x_i) \int \frac{\prod_k e^{-u_k^{(i)}}}{\mathcal{d}[\mathcal{U}(y_i, \zeta_i)]} \mathbf{1}(u^{(i)} \in \mathcal{U}_{y_i, \zeta_i}) \phi_{TN}(\zeta_i^* | u^{(i)}) \mathbf{1}(u^{(i)} \in \mathcal{U}_{y_i, \zeta_i^*}) du^{(i)} \frac{\mathcal{d}[\mathcal{U}(y_i, \zeta_i)]}{\mathcal{d}[\mathcal{U}(y_i, \zeta_i^*)]} \\ & \propto \exp(-0.5 \|\zeta_i - \beta x_i\|_2^2) \exp(-0.5 \|\zeta_i^* - \beta x_i\|_2^2) \\ & \quad \times \left(\prod_{j=1}^d \{ \Phi[(UA)_{i,j} - \beta_j^\top x_i] \mathbf{1}(y_{i,j} = 0) + \Phi[\beta_j^\top x_i - (UA)_{i,j}] \mathbf{1}(y_{i,j} = 1) \}^{-1} \right. \\ & \quad \left. \times \left[\prod_{k: (Ay_i - b)_k = 0} \exp(-u_k^{(i)}) \mathbf{1}(u^{(i)} \in \mathcal{U}(y_i, \zeta_i^*)) \mathbf{1}(u^{(i)} \in \mathcal{U}(y_i, \zeta_i)) \right] du^{(i)} \frac{1}{\mathcal{d}[\mathcal{U}(y_i, \zeta_i^*)]} \right), \end{aligned}$$

where we use ϕ_{TN} to denote the truncated normal density. Since $\mathcal{U}(y_i, \zeta_i)$ is invariant to the ordering between ζ_i and ζ_i^* , we see the backward joint probability kernel from ζ_i^* to ζ_i is exactly the same. Therefore, the Metropolis-Hastings acceptance ratio is one.

Remark 2 .

At a relatively high dimension d , the large difference between $\mathcal{U}(y_i, \zeta_i)$ and $\mathcal{U}(y_i, \zeta_i^*)$ may cause $P[u^{(i)*} \in \mathcal{U}(y_i, \zeta_i)] \approx 0$. To avoid such an issue, one can update a subvector of ζ_i for all i in each iteration. Empirically, we find that updating a subvector of length ≤ 100 keeps $P[u^{(i)*} \in \mathcal{U}(y_i, \zeta_i)]$ away from zero. For high dimensional problems, the above algorithm can lead to slow mixing performance; therefore, we present an alternative No-U-Turn Hamiltonian Monte Carlo (HMC) algorithm in the Supplementary Materials S5.

For drawing $u^{(i)}$ from (9), when $u^{(i)}$ is low-dimensional, we can use rejection sampling. When $u^{(i)*}$ is relatively high-dimensional, we use the hit-and-run algorithm (Smith, 1984). Specifically, with $u^{(i)}$ initialized at $\omega \in \mathcal{U}(y_i, \zeta_i)$: (i) draw a uniform direction $v: \|v\|_2 = 1$, subject to

$$\sum_{k: (Ay_i - b)_k = 0} v_k > 0 \text{ and } v_k = 0: (Ay_i - b)_k < 0; \text{ (ii) find the longest interval } (\tilde{a}, \tilde{b}) \text{ such that}$$

$$\omega + \alpha v \in \mathcal{U}(y_i, \zeta_i) \text{ for all } \alpha \in (\tilde{a}, \tilde{b}), \text{ then draw } \alpha^* \sim \text{Exp}_{(\tilde{a}, \tilde{b})} \left(\sum_{k: (Ay_i - b)_k = 0} v_k \right), \text{ which is a truncated}$$

exponential distribution, with rate $\sum_{k: (Ay_i - b)_k = 0} v_k$ and supported in $(\tilde{a}, \tilde{b})$; (iii) update ω to $\omega + \alpha^* v$

. We repeat (i)-(iii) 100 times. The hit-and-run algorithm is efficient and shown to mix rapidly for drawing samples in a polyhedron (Lovász and Vempala, 2006; Chen and Eldan, 2022). In

case one has an unbounded polyhedron, one can truncate $\tilde{a}$ and $\tilde{b}$ at a large magnitude such as 10^5 . We see that (9) is always proper regardless if $\mathcal{U}(y_i, \zeta_i)$ is bounded or not.

Remark 3 .

When hit-and-run is used to draw $u^{(i)}$, we effectively draw from an approximate distribution hence making the algorithm approximate MCMC (Johndrow et al., 2015).

In the Supplementary Materials S5, we present two alternative samplers, the HMC and pseudo-marginal sampler, and show their computational performance in simulation.

4 Consistency theory

This section develops posterior consistency theory for Bayesian inference on combinatorial response models. We establish conditions under which the posterior distribution concentrates around the true parameter as sample size grows. We consider a setting in which the response variables Y_i take values in the vertex set $\mathcal{E} \subseteq \{0,1\}^d$ of an integral polytope

$\mathcal{P} = \{z \in [0,1]^d : Az \leq b\}$, with d fixed and finite, while the sample size n diverges. We focus on the following two models:

$$\text{Intercept-only model: } Y_i = T(\zeta_i), \quad \zeta_i \mid \mu \stackrel{i.i.d.}{\sim} N(\mu, I_d), \quad \mu \sim \pi, \quad \forall i = 1, \dots, n. \quad (10)$$

$$\text{Regression model: } Y_i = T(\zeta_i), \quad \zeta_i \mid \beta \stackrel{\text{indep}}{\sim} N(\beta x_i, I_d), \quad \beta \sim \pi, \quad \forall i = 1, \dots, n. \quad (11)$$

where $T : \mathbb{R}^d \setminus C \rightarrow \mathcal{E}$ is the argmax mapping $T(\zeta) = \arg\max_{e \in \mathcal{E}} \zeta^\top e$, and π represents the prior distribution on either μ or β , depending on the model under consideration.

To simplify notation, we denote by $Y^n := (Y_1, \dots, Y_n)$ the response data and by P_μ^n the joint distribution of Y^n given μ , with

$$P_\mu(e) := P(Y = e \mid \mu) = \int_{\mathbb{R}^d} \phi(\zeta; \mu, I_d) \mathbb{1}(\zeta^\top e = \max_{e' \in \mathcal{E}} \zeta^\top e') d\zeta, \quad \forall e \in \mathcal{E}.$$

4.1 Inconsistency of unconstrained model

For combinatorial responses, it might appear tempting to simply ignore the constraints and model each data point as an array of independent discrete outcomes. However, we show this model will lead to inconsistent estimates.

We illustrate the inconsistency using intercept-only models. Suppose the ground-truth data generating distribution is $P^0(Y = e) = p_e^0$ for each combinatorial outcome $e \in \{0,1\}^d \cap \mathcal{P}$, and we

use $q_{j1} = \sum_{\text{all } e} p_e^0 1(e_j = 1)$ to denote the marginal $P^0(Y_j = 1)$. We consider the case that each $q_{j1} \in (0, 1)$, so that both 1 and 0 can appear in j -th dimension with positive probability.

Consider an unconstrained model that ignores the constraint $Az \leq b$, and hence models the responses as $Y_{ij} \stackrel{\text{indep}}{\sim} \text{Bernoulli}[\Phi(\mu_j)]$, with a prior $\mu \sim \pi(\mu)$. The resulting likelihood-prior product is

$$\exp \left[n \sum_{j=1}^d \left\{ \frac{n_{j1}}{n} \log \Phi(\mu_j) + \frac{n - n_{j1}}{n} \log \Phi(-\mu_j) \right\} + \log \pi(\mu) \right],$$

where $n_{j1} = \sum_{i=1}^n \mathbf{1}(y_{ij} = 1)$ denotes the number of observed ones at the j -th coordinate. Assuming the prior $\pi(\mu)$ has full support on $\mathbb{R}^d$ and is invariant to n , the log-likelihood term dominates as $n \rightarrow \infty$. Consequently, the posterior for $\Phi(\mu_j)$ converges almost surely to a point mass:

$\Pi(\Phi(\mu_j) | y_1, \dots, y_n) \xrightarrow{a.s.} \delta_{q_{j1}}$, where δ denotes a Dirac measure; this is analogous to the way that the maximum likelihood estimator (MLE) $\hat{\Phi}(\mu_j) = n_{j1} / n$.

Consequently, the fitted value under the unconstrained model has:

$$\tilde{P}(Y = e) \xrightarrow{a.s.} \prod_{j=1}^d q_{j1}^{e_j} (1 - q_{j1})^{1-e_j}.$$

Note that the limiting law of Y above is not guaranteed to be the same as p_e^0 . Perhaps the most striking difference is that for a value outside the combinatorial space, $e_1 \in \{0, 1\}^d \setminus \mathcal{P}$, we have $\tilde{P}(Y = e_1) > 0$, but the ground-truth is $P^0(Y = e_1) = 0$. For the example in Section 2.1, if we have the ground truth $\mu_0 = 0$, then $n_{11} / n \xrightarrow{a.s.} 3/8$, $n_{21} / n \xrightarrow{a.s.} 3/8$, hence $\tilde{P}[Y = (1, 1)] \xrightarrow{a.s.} 9/64$, but we know $Y = (1, 1)$ cannot happen due to the constraint.

Remark 4.

In the above exposition, we focus on the unconstrained models with conditional independence across the dimensions of y_i . As an alternative, an unconstrained model could potentially avoid the inconsistency issue if one considers a latent correlated $\tilde{\zeta}_i \sim N(\mu, \Omega^{-1})$ and $y_{i,j} = 1(\tilde{\zeta}_{i,j} > 0)$, where Ω characterizes the conditional dependence structure of y_i . On the other hand, a caveat of such a latent Gaussian model is that ignoring the combinatorial constraint could lead to misleading conditional dependence estimate, for two outcomes that are marginally independent. For example, consider three binary outcomes, $y_{i,1} \sim \text{Bernoulli}(0.5)$, $y_{i,3} \sim \text{Bernoulli}(0.5)$, $y_{i,1}$

and $y_{i,3}$ are independent; $y_{i,2}$ is subject to combinatorial constraint $y_{i,2} = 0$ if $y_{i,1} = 0$, otherwise $y_{i,2} \sim \text{Bernoulli}[(y_{i,3} + 1) / 3]$. Now $y_{i,1}$ and $y_{i,3}$ become conditionally dependent given $y_{i,2}$ due to the combinatorial constraint.

4.2 Consistency under inverse integer linear program link

We now discuss the consistency results. We focus on the scenario when the data generating distribution is (10) or (11), with ground truth μ_0 or β_0 . Our exposition progresses over three stages: (i) the consistent recovery of the law P_{μ_0} under the intercept-only model, regardless of whether we can recover μ_0 ; (ii) the consistent recovery of μ_0 for the intercept-only model; (iii) the consistent recovery of β_0 in the regression model. Although (ii) may be viewed as a special case of (iii), establishing posterior consistency in regression is much more challenging than in the intercept-only model because the data are no longer identically distributed. Therefore, we keep (ii) and (iii) as two separate results.

4.2.1 Recovering probability law via intercept-only model

We first focus on recovering the ground-truth law of Y , $P(Y|\mu_0)$, under an intercept-only generative distribution. For recovering the above, there is not much requirement needed for posterior consistency other than prior support, as described below.

Condition 1 (Prior support).

The prior π is a probability measure on the parameter space ($\mathbb{R}^d$ for the intercept-only model or $\mathbb{R}^{d \times p}$ for the regression model) satisfying $\pi(B_\varepsilon(\theta_0)) > 0$ for every open ball $B_\varepsilon(\theta_0)$ centered at the true value of the parameter θ_0 .

Condition 1 ensures that the prior places positive mass in every Kullback–Leibler (KL) neighborhood of the true parameter. This is the usual KL support requirement for posterior consistency (Schwartz, 1965) and is satisfied by commonly used continuous priors.

Theorem 4 .

Under Condition 1, the posterior of the intercept-only model (10) is strongly consistent at estimating P_{μ_0} . That is, $\left\| \int P(Y|\mu) \Pi(\mu|Y^n) d\mu - P(Y|\mu_0) \right\|_1 \xrightarrow{n \rightarrow \infty} 0$ $P_{\mu_0}^\infty$ – almost surely.

4.2.2 Recovering coefficients in intercept-only model

We want to clarify that the consistent recovery for P_{μ_0} does not imply the recovery for μ_0 . An identifiability issue may arise. For example, when the response is a one-hot vector determined by

the Gaussian latent variable that attains the maximum value:

$$y_i = \operatorname{argmax}_z \zeta_i^\top z \quad \text{subject to } z \in \{0,1\}^d \text{ and } \sum_{j=1}^d z_j = 1,$$

shifting all coordinates of μ by a constant scalar does not change the probability distribution of the outcomes. For such cases, a common strategy to ensure identifiability in generalized linear model is to impose equality constraint on the parameter μ , such as setting $\mu_1 = 0$ — equivalently, one assigns a degenerate prior $\pi(\mu)$.

On the other hand, there are cases under which μ is identifiable without the need of degenerate prior. It turns out that the geometry of the polytope $\mathcal{P}$ plays a key role in ensuring identifiability of μ . In fact, we show that the non-degeneracy of the polytope $\mathcal{P}$ is equivalent to the identifiability of μ in Theorem 5.

Condition 2 (Polytope structure).

The polytope $\mathcal{P} = \{z \in \mathbb{R}^d : Az \leq b\} \subset \mathbb{R}^d$ has nonempty interior in $\mathbb{R}^d$.

Theorem 5 (Intercept-only model identifiability).

Consider the intercept-only model (4). For any $\mu, \mu' \in \mathbb{R}^d$,

$$P_\mu = P_{\mu'} \quad \text{if and only if} \quad \mu - \mu' \in \operatorname{Span}\{e - e' : e, e' \in \mathcal{E}\}^\perp. \quad (12)$$

Consequently, the model (4) is identifiable (i.e., P_μ is injective in μ) if and only if Condition 2 holds. When Condition 2 fails, the model is identifiable only up to shifts in directions orthogonal to $\operatorname{Span}\{e - e' : e, e' \in \mathcal{E}\}$.

A commonly used approach for establishing posterior consistency is Schwartz's Theorem (Schwartz, 1965), which involves the construction of an exponentially consistent sequence of tests as introduced below.

Definition 1.

Let $V^c \subset \mathbb{R}^d$ and $\mu_0 \in \mathbb{R}^d$. A sequence of tests $\psi_n(Y^n) : \mathcal{E}^n \rightarrow [0,1]$ is said to be *exponentially consistent* for testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \in V^c$, if there exists $C_1, C_2, b > 0$ such that

$$(a) \quad \mathbb{E}_{P_{\mu_0}^n} [\psi_n] \leq C_1 e^{-nb};$$

$$(b) \quad \inf_{\mu \in V^c} \mathbb{E}_{P_\mu^n} [\psi_n] \geq 1 - C_2 e^{-nb}.$$

With these ingredients, we now establish the posterior consistency of the intercept-only model for recovering $\mu_0 \in \mathbb{R}^d$.

Theorem 6 (Posterior consistency for the intercept-only model).

Under Conditions 1 and 2, the posterior of the model (10) is consistent at estimating μ_0 , i.e., for any open neighborhood of μ_0 , denoted by V_{μ_0} , the following holds:

$$\Pi(V_{\mu_0}^c | Y^n) \xrightarrow{n \rightarrow \infty} 0 \quad P_{\mu_0}^\infty - \text{almost surely.}$$

4.2.3 Recovering coefficients in regression model

We now establish posterior consistency for the regression model (11). Following Amewou-Atisso et al. (2003) and Sriram et al. (2013), we treat the covariates x_i as fixed. A key distinction from the intercept-only model is that the observations $Y_1, \dots, Y_n$ are independent but not identically distributed, since the distribution of each Y_i depends on the fixed covariate x_i . Let $Q_{i,\beta}$ denote the conditional law of Y_i given βx_i . Let Q_β^n denote the conditional law of $Y^n = (Y_1, \dots, Y_n)$ given $\{\beta x_i\}_{i=1}^n$. We have $Q_\beta^n = \prod_{i=1}^n Q_{i,\beta}$. We have the posterior distribution of β given Y^n , for any π -measurable set $\mathcal{B} \subset \mathbb{R}^{d \times p}$:

$$\Pi(\mathcal{B} | Y_1, \dots, Y_n) = \frac{\int_{\mathcal{B}} Q_\beta^n(Y_1, \dots, Y_n) d\pi(\beta)}{\int_{\mathbb{R}^{p \times d}} Q_\beta^n(Y_1, \dots, Y_n) d\pi(\beta)}.$$

For two discrete probability distributions G and G' on a common support $\mathcal{E}_0$, we define the KL divergence $\text{KL}(G \| G') := \sum_{e \in \mathcal{E}_0} G(e) \log(G(e) / G'(e))$, and the associated second-moment quantity $V(F, G) := \sum_{e \in \mathcal{E}_0} G(e) [0 \vee \log(G(e) / G'(e))]^2$.

With the above notations, we now employ an adapted version of Schwartz's Theorem showed by Amewou-Atisso et al. (2003). For this article to be self-contained, we first state the theorem below using our notations.

Theorem 7 .

(Amewou-Atisso et al., 2003) *Suppose the following conditions hold:*

- (i) *There exists an exponentially consistent sequence of tests $\psi_n(Y^n) : \mathcal{E}^n \rightarrow [0, 1]$ for testing $H_0 : \beta = \beta_0$ vs. $H_1 : \beta \in V_{\beta_0}^c$. That is, there exist constants $C_1, C_2, b > 0$:*

$$(a) \mathbb{E}_{Q_{\beta_0}^n} [\psi_n] \leq C_1 e^{-nb};$$

$$(b) \inf_{\beta \in V_{\beta_0}^c} \mathbb{E}_{Q_{\beta}^n} [\psi_n] \geq 1 - C_2 e^{-nb}.$$

(ii) For every $\delta_0 > 0$,

$$\pi \left(\left\{ \beta : \sup_i KL(Q_{i,\beta_0} \| Q_{i,\beta}) < \delta_0, \sum_{i=1}^{\infty} \frac{V(Q_{i,\beta_0}, Q_{i,\beta})}{i^2} < \infty \right\} \right) > 0.$$

Then, $\Pi(V_{\beta_0}^c | Y^n) \xrightarrow{n \rightarrow \infty} 0$ $Q_{\beta_0}^{\infty}$ - almost surely.

Condition (ii) in Theorem 7 is the adapted KL-support condition for the non-i.i.d. case, which requires Condition 1, and the boundedness of covariates (Condition 3). The latter condition is ubiquitously assumed in establishing posterior consistency for regression models, and is needed to bound the supremum of the Kullback-Leibler divergence.

Condition 3 (Boundedness of Covariates).

There exists $\ell > 0$ such that $\sup_i \|x_i\|_2 \leq \ell$.

To distinguish $\beta \neq \beta_0$ from β_0 , the covariates must satisfy additional directional condition. We introduce the following definition, and then present the additional condition.

Definition 2 (Face selection mapping).

For a polyhedron $\mathcal{P}$, a face is a subset $F = \{z \in \mathcal{P} : v^\top z = \max_{z \in \mathcal{P}} v^\top z\}$ for some $v \in \mathbb{R}^d$. A *proper face* is one that is not equal to $\mathcal{P}$. We define *face selection mapping* as:

$$\mathcal{F}(v) := \{e^* \in \mathcal{P} : v^\top e^* = \max_{e \in \mathcal{P}} v^\top e\}.$$

We note that a face can be an empty set, a vertex, a line segment, or a higher-dimensional subset of $\mathcal{P}$. Our primary focus below is on $\mathcal{F}(v)$ that could yield a vertex subset in $\mathcal{E}$.

Condition 4 (Covariates richness).

For every $u \in \mathbb{S}^{dp-1} := \{\beta \in \mathbb{R}^{d \times p} : \|\beta\|_F = 1\}$ where $\|\cdot\|_F$ denotes the Frobenius norm, there exist a proper face set $F_u : \mathcal{E} \cap F_u \subsetneq \mathcal{E}$ and a constant $\delta_u > 0$, such that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\mathcal{F}(ux_i) = F_u, \min_{e \in \mathcal{E} \cap F_u, e' \in \mathcal{E} \setminus F_u} (e - e')^\top ux_i \geq \delta_u\} > 0.$$

Remark 5 .

Intuitively, Condition 4 requires that for every unit direction u in the parameter space, there exists a nontrivial subset of covariates along which the perturbation $\beta_0 + u$ produces a detectable separation between $e^\top u x_i$ and $e'^\top u x_i$ for $e \in \mathcal{E} \cap F_u$ and $e' \in \mathcal{E} \setminus F_u$. This ensures that any deviations from β_0 can be detected by a subset $\mathcal{X}$ of covariates such that $|\mathcal{X} \cap \{x_i : i \in [n]\}|/n$ is not vanishing as n diverges.

Remark 6 .

Condition 4 is an essentially minimal requirement for posterior consistency in our regression model. When it fails, $\beta_0 + u_0$ and β_0 are distinguishable only on an increasingly smaller proportion of covariates as n diverges, since there exists a direction $u_0 \in \mathbb{S}^{dp-1}$ such that

$\liminf_{n \rightarrow \infty} (1/n) \sum_{i=1}^n 1\{\mathcal{F}(u_0 x_i) = F, \min_{e \in \mathcal{E} \cap F, e' \in \mathcal{E} \setminus F} (e - e')^\top u_0 x_i \geq \delta\} = 0$ for every proper face set F and every $\delta > 0$.

Theorem 8 (Posterior consistency for the regression model).

Under Conditions 1, 3, and 4, the posterior of the model (11) is consistent at estimating β_0 , i.e., for any open neighborhood of β_0 , denoted by V_{β_0} , the following holds:

$$\Pi(V_{\beta_0}^c | Y^n) \xrightarrow{n \rightarrow \infty} 0 \quad Q_{\beta_0}^\infty - \text{almost surely.}$$

Remark 7 .

Throughout this section we treat the covariates $\{x_i\}_{i \geq 1}$ as fixed. Nevertheless, Condition 4 has a natural interpretation under random design. Suppose that x_i are i.i.d. draws from a continuous distribution P_X supported on a compact subset of $\mathbb{R}^p$. If the empirical frequencies in Condition 4 are replaced by probabilities under P_X , namely if for every $u \in \mathbb{S}^{dp-1}$ there exist a proper face set F_u such that $\mathcal{E} \cap F_u \subsetneq \mathcal{E}$ and $\delta_u > 0$ such that

$$P_X \left(\mathcal{F}(uX) = F_u, \min_{e \in \mathcal{E} \cap F_u, e' \in \mathcal{E} \setminus F_u} (e - e')^\top uX \geq \delta_u \right) > 0,$$

then Condition 4 holds almost surely by the law of large numbers. In this case, the posterior consistency result in Theorem 8 is expected to remain valid under random design.

5 Simulation studies

5.1 Fitted value curve

We first conduct a simple simulation to demonstrate the importance of incorporating combinatorial constraints in the model. We revisit the example from Section 2.1 where responses $y_i \in \{0,1\}^2$ must satisfy the constraint $y_{i,1} + y_{i,2} \leq 1$, meaning at most one component can be 1.

The data generating process follows our latent variable framework where $\zeta_i \sim N(\mu_i, I_2)$ and the mean μ_i has a linear structure $\mu_i = \beta_0 + \beta_1 x_i$. Here $\beta_0, \beta_1 \in \mathbb{R}^2$ are coefficient vectors and $x_i \in \mathbb{R}$ is a scalar covariate. We use a one-dimensional covariate x_i in order to clearly illustrate the fitted value curves.

We compare two models: (1) a standard bivariate probit regression that ignores the constraint (unconstrained model), and (2) our proposed model that incorporates the constraint (constrained model). Figure 2 shows the fitted value curves, as the estimated class probabilities from both models alongside the ground truth probabilities computed using the true coefficients β_0 and β_1 .

Figure 2(b,c) show that our model successfully recovers the true probability distribution. On the other hand, panel (a) shows that the unconstrained model produces biased probability estimates. Clearly, the unconstrained model incorrectly assigns positive probability to the outcome (1,1) that should have had probability zero.

A natural question is if one could correct the bias of the unconstrained model by post-processing the predicted probabilities via conditioning on the constraint. Unfortunately, the answer is negative. As shown in Figure 2(a), at $x_i \approx -1.4$, the unconstrained model incorrectly assigns almost equal probabilities to the three outcomes (0,0), (1,0), and (0,1), whereas the ground truth probability of (0,0) is close to zero.

5.2 Signal recovery and computational efficiency

We evaluate the performance of our method with simulation mimicking skip-logic surveys. To illustrate, we first consider a short survey consisting of $d = 5$ binary questions, where later questions are conditionally presented based on earlier responses. For instance, respondents are first asked a screening question (Question 1), and only those answering positively are presented with follow-up questions (Questions 2 and 3). Additional questions (Questions 4 and 5) are shown only if specific combinations of earlier responses are observed. As a result, certain response patterns are infeasible by design, and show as structural zeros in the response vector. Table 1 shows a short online health survey for studying preventive care behavior and its skip-logic conditions.

This skip logic can be encoded using partial order constraints on the response vector. For the example in Table 1, we can use $y_{i,2} \leq y_{i,1}, y_{i,3} \leq y_{i,1}, y_{i,4} \leq y_{i,2}, y_{i,5} \leq y_{i,4}$, which corresponds to

$Az \leq 0$ ($A \in \{-1, 0, 1\}^{4 \times 5}$), where each row of A has exactly one -1 and one 1 . Equivalently, the matrix A can be viewed as the directed incidence matrix of a directed graph, hence guaranteed to be TUM.

Our simulation study adopts this skip-logic survey mechanism to form constraints and simulate data over different dimensions. Specifically, we generate a random graph containing m nodes and d edges, and use its directed incidence matrix as A . We use a binary vector for b . We simulate data from a regression model with a true coefficient matrix $\beta \in \mathbb{R}^{d \times p}$. Each simulation uses $n = 1000$, $p = 5$ and d from a varying range. We examine how the method scales with response dimension by considering nine different values of d , ranging from small ($d = 2, 5, 10, 20, 50$) to large ($d = 100, 200, 500, 1000$). The number of constraints m is chosen from $\{1, 5, 10, 20, 50, 100\}$. For prior specification, we assign $\beta \sim \text{Mat-N}(0, I_d, I_p \tau)$ with τ set as 10.

For each simulation, we perform MCMC sampling with 50000 total iterations, with the first 5000 discarded as burn-in and retaining every 25th sample as thinning thereafter. The trace plots and autocorrelation function (ACF) plots shown in Figure 3 indicate efficient mixing of the Markov chain, with quick decay in the autocorrelation structure.

For the running time per 1000 iterations, it takes 0.07 minutes for $(d, m) = (2, 1)$, 8 minutes for $(d, m) = (1000, 1)$, and 52 minutes for $(d, m) = (1000, 100)$, using Rcpp on MacBook Pro with a 8-core CPU. R source code is given in the Supplementary Materials.

To assess estimation accuracy, we calculate the root mean squared error (RMSE) between the posterior mean estimates and the ground-truth values, presented in Table 2. Figure 4 displays the posterior distribution for each $\beta_{i,j}$ parameter, demonstrating that the credible intervals effectively capture the true parameter values. We provide additional details in the Supplementary Materials S4.

6 Data analysis on waterfowl matching

We conduct data analysis on the seasonal matching of waterfowl. All waterfowl are monogamous, though their mating patterns differ across species. Geese, swans, and whistling ducks exhibit perennial monogamy with lifelong pair bonds, while ducks practice seasonal monogamy where pair bonds last 4 to 8 months, and they choose new mates each year. The pair formation process follows a clear seasonal pattern: pairing initiates during fall migration in September–October, most pairs are formed by December–January, and pair bonds continue to strengthen through winter and spring.

Our dataset is the temporal observation of the pairing of male and female ducks from November to March. In the dataset, there are 95 ducks ($|V| = 95$) in total, and from 7 different species: American Black Duck, Mallard, Gadwall, Scaup, Redhead, Canvasback, and Ring-necked Duck. Since pairs can only be formed between a male and a female within the same species, the

observed matching $M^t = (V, E_M^t)$ at time t is the subgraph of a bipartite graph $G = (V, E_G)$, with 339 possible edges ($|E_G| = 339$). As described in Example 4 of Section 2, we use A as the incidence matrix for G , and convert each M^t into a binary vector $y_t \in \{0,1\}^d$ with $d = 339$, corresponding to whether the e -th edge in G is chosen into M^t . The polyhedron associated with matching is then defined by $Ay_t \leq 1$. The maximal number of ones in y_t is 46, which is substantially smaller than 339 due to the polyhedron constraint.

For the covariates, time is modeled using B-splines corresponding to weeks. We use B-spline bases corresponding to $\kappa = 5$ columns in the spline matrix. Additionally, we incorporate the weights of both male and female ducks measured at the end of November. Our model for each latent variable $\zeta_t \in \mathbb{R}^d$ is $\zeta_t = \beta_0 \mathbf{1}_d + x_{\text{weights}} \beta_1 + C_{\text{species}} \beta_2 x_{\text{splines},t} + \epsilon_t$, $\epsilon_t \sim N(0, I_d)$ where $\beta_0 \in \mathbb{R}$ is the common intercept, $x_{\text{weights}} \in \mathbb{R}^{d \times 2}$ has each row as the weights of two ducks on each edge of G , $\beta_1 \in \mathbb{R}^2$ is the coefficient vector for duck weights, $x_{\text{splines},t} \in \mathbb{R}^{\kappa \times 1}$ corresponds to the B-spline basis function value at time t , $\beta_2 \in \mathbb{R}^{7 \times \kappa}$ is the coefficient matrix corresponding to 7 individual curves for species, $C_{\text{species}} \in \{0,1\}^{d \times 7}$ is the indicator matrix for species in which each row has only one element equal one. For prior specification, we assign $\beta_0, \beta_{1:1}, \beta_{1:2} \stackrel{i.i.d.}{\sim} N(0, \tau_\alpha^{-1})$ and $\beta_{2:(i,\cdot)} \stackrel{indep}{\sim} N(0, (\tau_\beta s_i)^{-1} I_\kappa)$ with s_i denoting the number of ducks of species i . We use the parameterization with s_i in order for the full conditional of β_2 to be a matrix normal distribution. We set $\tau_\alpha = \tau_\beta = 0.1$. The posterior sampling method can be found in the Supplementary Materials S2.

We run 50000 MCMC iterations, with the first 2000 treated as burn-in and the rest thinned at every 20th iteration. Figure 5 and 6 show the trace plots for four randomly selected parameters and the ACF plots, which show excellent mixing. Table 3 presents the posterior means and 95% credible intervals for the coefficient estimates of $\beta_{1:1}$ and $\beta_{1:2}$, corresponding to the fixed effects of the male and female duck weights, respectively. The effect of female duck weight is statistically significant, while male duck weight is not.

We now show how the matching probability changes over time for each species. We plot the probability curves separately for male and female ducks, while fixing the weight to be the medium of the males and females, respectively. The probabilities are computed by Monte Carlo integration, which involves sampling 1000 ζ_t 's given β at each time point t and computing the associated $\tilde{y}_t = T(\zeta_t)$ using linear program. We evaluate the probability of $\sum_{e \in (i, j^*) \text{ or } (j^*, i)} \tilde{y}_{t,e} = 1$ (j^* is the index of the medium-weight duck), corresponding to the event that the duck j^* forms a match with another one. We calculate these probabilities for each Markov chain sample of β , allowing us to obtain point-wise 95% credible bands. We plot the curves in Figure 7. In addition, we overlay the mean curves between the two sexes and show comparison in the Supplementary Materials S3.

We can see a clear difference between dabbling ducks (American Black Duck, Mallard, Gadwall) and diving ducks (Scaup, Redhead, Canvasback, Ring-necked Duck). This is consistent with the biological knowledge that dabbling ducks tend to form pairs much earlier than diving ducks. On the other hand, we can see that Gadwall form matchings slightly later than American Black Duck and Mallard in the year.

Next, given that there are more females than males for some species in this dataset, we assess the effects of competition on the matching probabilities. To be exact, we compute the conditional probability given that there is at least one opposite-sex duck of the same species that is not matched to any other duck, and view it as a surrogate measure of matching probability if there were no competition. We provide the result in Figure 8. For Mallard, the effects of competition start to show as early as in the fourth week, whereas the effects show for Redhead and Scaup later in the season.

7 Discussion

In this article, we propose a likelihood-based approach for handling combinatorial response data. We take advantage of the properties of integer linear program, which lead to a data augmentation strategy that enjoys both tractable theoretic analysis and efficient computation. There are several interesting directions worth further pursuing. First, for consistency theory, we consider the case when the ground truth is covered by the family of our specified models. On the other hand, it remains to be seen if one could recover the fitted value functional under some level of model misspecification. Second, we could extend the strategy of augmenting observed data with optimization decision variables beyond linear program. For example, there is a class of optimization approaches based on semi-definite program, which are applicable to inference tasks such as covariance modeling and metric learning.

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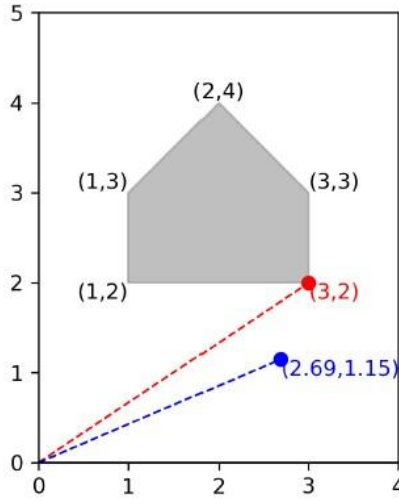
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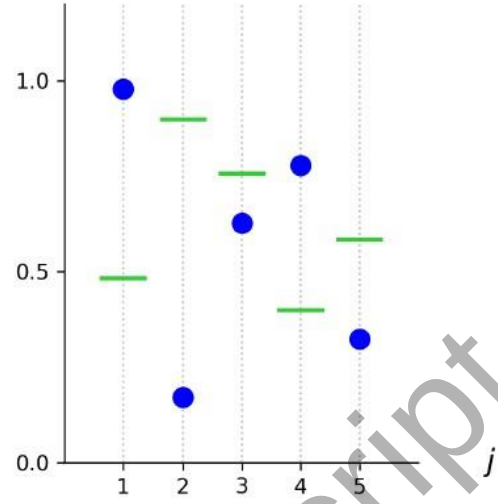
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(a) Solving a continuous linear program in an integral polyhedron (gray, whose vertices are all integer-valued) is equivalent to transforming a continuous coefficient vector ζ_i (blue) into an integer coordinate y_i (red).



(b) When y_i is binary, there is a dual thresholding representation corresponding to whether $\zeta_{i,j}$ (blue) is greater or equal to $A'_{j,j}u^{(i)}$ (green bar), with $u^{(i)}$ the dual variable. In the figure, the associated y_i is $(1, 0, 0, 1, 0)$.

Figure 1: Illustration of using the linear program as a mapping (Panel a), and its equivalent representation when the output is restricted to be binary (Panel b).

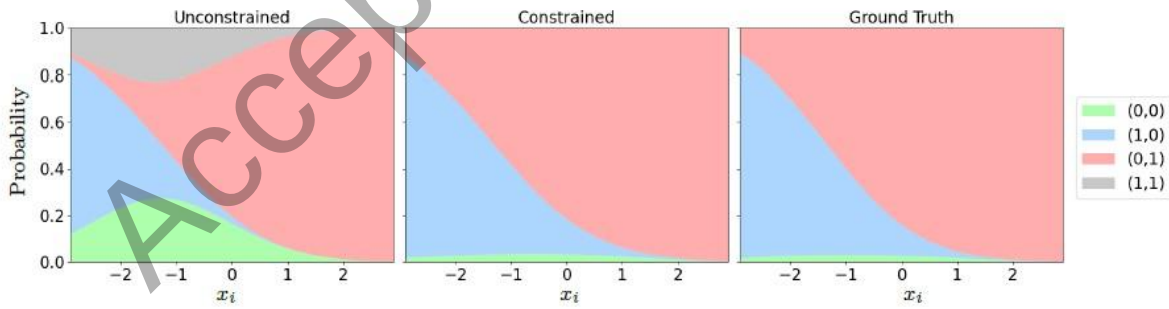


Figure 2: Comparison of predicted class probabilities under unconstrained and constrained models. Each panel shows the stacked predicted probabilities for the four response patterns as a function of the covariate x_i .

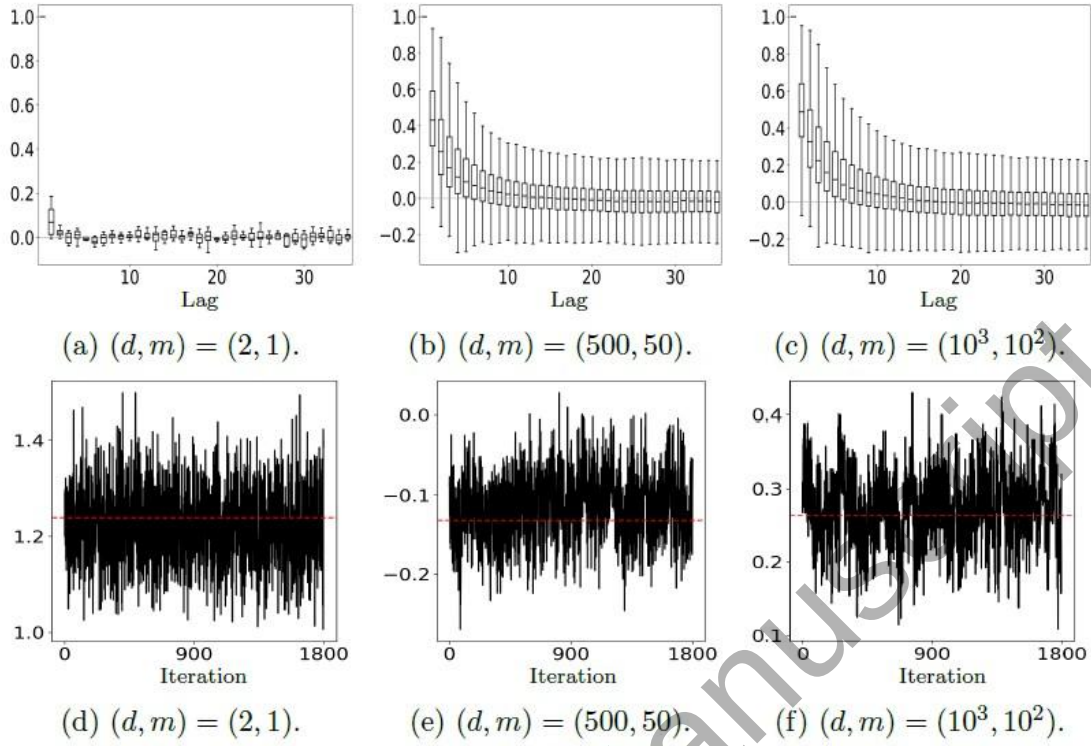


Figure 3: ACF plots and trace plots for different values of (d, m) . Each box displays the ACF of all $\beta_{i,j}$ parameters. The ACF is computed after thinning. The trace plots are based on a randomly chosen parameter under each setting.

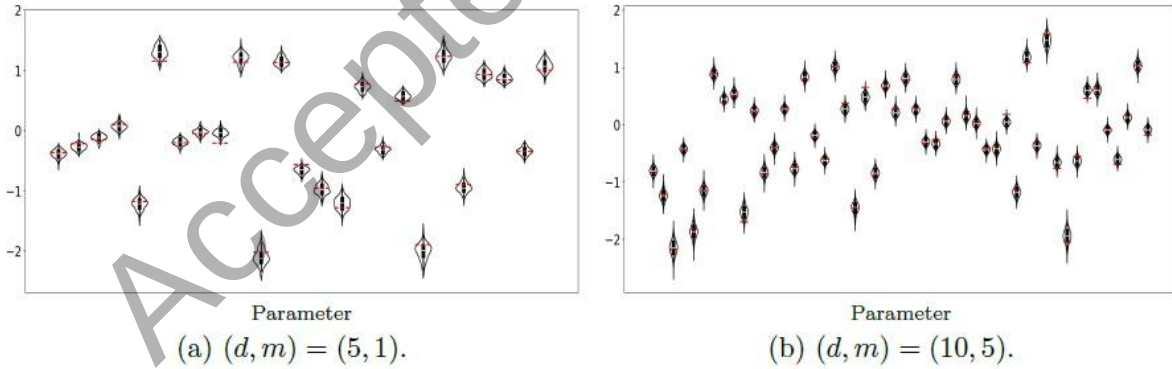


Figure 4: Violin plots of the posterior distributions of $\beta_{i,j}$ from MCMC sampling for different values of (d, m) . The horizontal lines represent the ground truth.

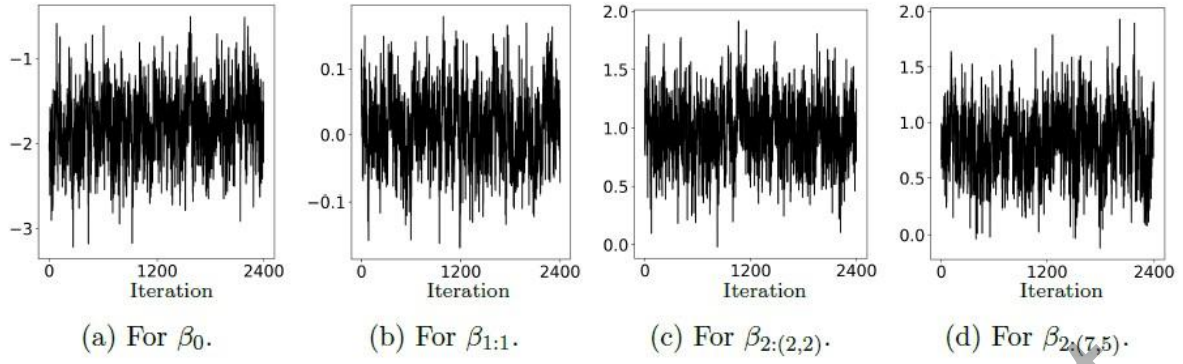


Figure 5: Trace plots of randomly selected parameters for the duck matching model.

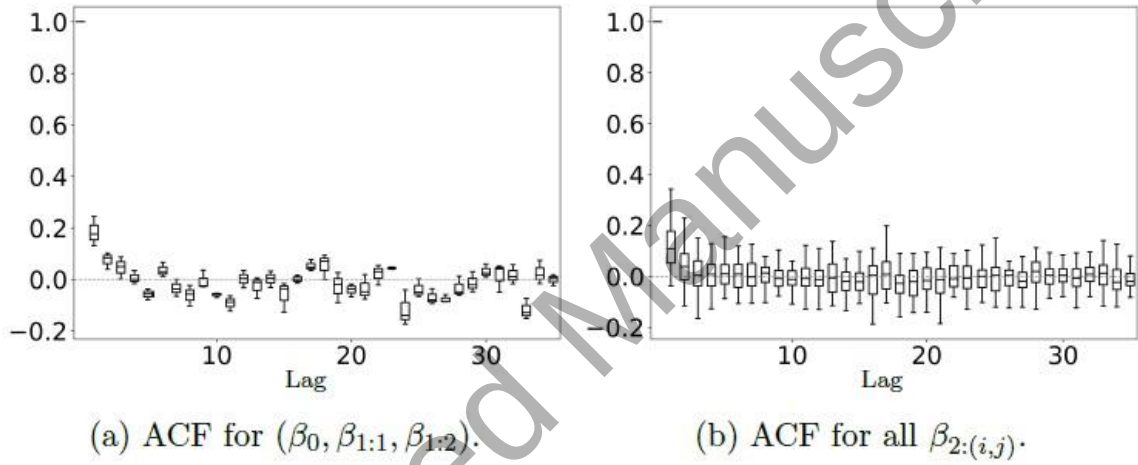


Figure 6: ACF plots for the duck matching model. Each box displays the ACFs of all parameter components.

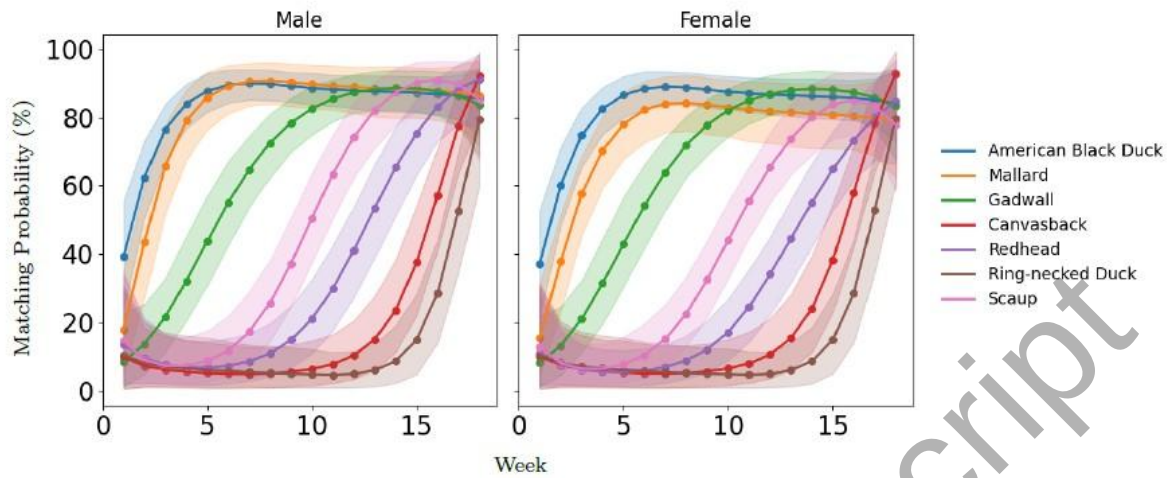


Figure 7: Matching probabilities over time with pointwise 95% credible bands.

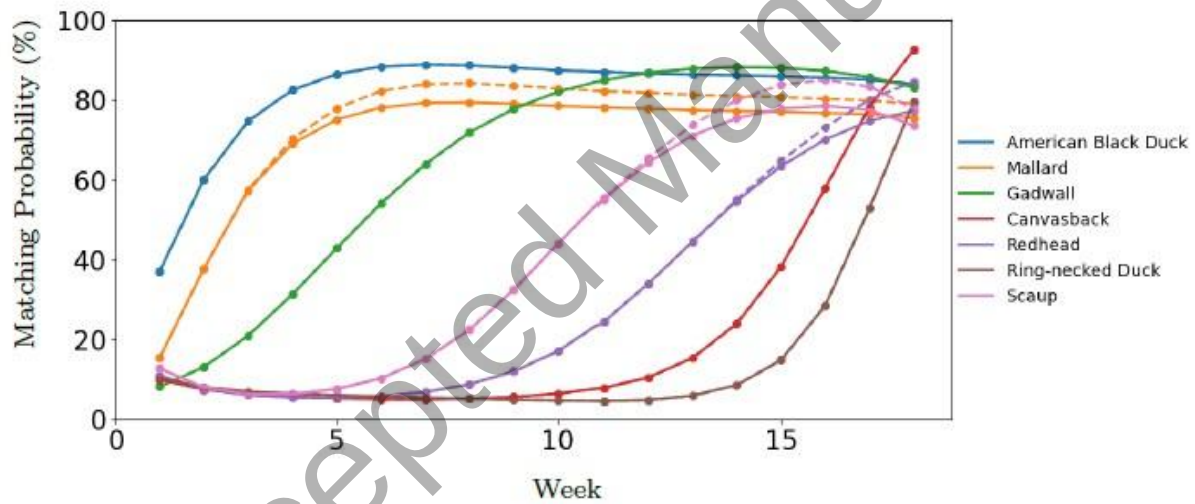


Figure 8: Mean matching probabilities over time for female ducks. Solid lines correspond to having competition, and dashed lines correspond to no competition.

Table 3: Coefficient estimates and 95% credible intervals for β_0 and β_1 .

Parameter	Estimate	Credible Interval
Intercept β_0	-1.786	(-2.566, -1.006)
Male weight $\beta_{1,1}$	0.010	(-0.090, 0.116)
Female weight $\beta_{1,2}$	0.187	(0.012, 0.362)

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